

Jens Laufer

Forward Deployed Engineer — autonomous AI agents and the harness that makes them reliable

Harness & Agentic Engineering · Forward Deployed · Python · LLM agents / MCP · RAG · Enterprise integration · Java / Spring Boot · Vue 3 · Docker

~29 years of software engineering (freelance since 1997), a second track in data science and machine learning — and since late 2025 harness and agentic engineering: the scaffolding around an LLM that turns a language model into a system you can leave running. Complex projects delivered agile and test-driven, always close to the client.

AVAILABLE from 15 September 2026	WORKS worldwide	REMOTE SHARE 95 % 5 % on-site
HOURLY RATE €100/h on-site €90/h remote · excl. VAT	BASED IN Karlstein am Main Germany	

FOCUS

- ~29 years of software engineering (since 1997), freelance throughout.
- Java/Spring Boot microservices with Vue.js and React front-ends — end-to-end with Docker, Kubernetes and GitLab CI.
- Strictly test-driven (TDD, JUnit, Mockito, Playwright/Selenium).
- Second track in data science and machine learning — Udacity ML Engineer & Data Analyst, Kaggle, writer at Towards Data Science.
- Broad industry exposure: banking/finance, insurance, automotive, telecommunications, travel, industry, retail, marketing and startups.

ROLES

- Full-stack engineering
- Solution design
- Backend engineering
- DevOps
- Build & configuration management
- Data analysis
- Prototyping
- Machine learning
- Jenkins plugin development
- Maven plugin development

SKILLS

LANGUAGES

- Java
- JavaScript
- TypeScript
- Python
- R
- Groovy

MICROSERVICES / BACKEND

- Spring Boot
- Spring
- Spring Data
- Spring Integration
- Spring Security
- OAuth2
- Microservices
- Feign
- Sleuth
- Zipkin
- Python Eve
- Python Flask
- Java EE
- EJB

FRONTEND

- Vue.js
- React
- AngularJS
- GWT
- Quasar
- HTML
- CSS
- Bootstrap

API / INTEGRATION

- REST
- Swagger
- OpenAPI
- SOAP
- Webservices
- CXF
- XFire
- Kafka
- ServiceMix

SECURITY / IAM

- Keycloak
- Spring Security
- OAuth2

MACHINE LEARNING

- scikit-learn
- Keras
- DeepLearning4J
- CNN
- Transfer Learning
- Supervised/Unsupervised
- k-means
- PCA
- t-SNE
- Reinforcement Learning
- OpenAI Gym
- Topic Modeling

TOOLING / OTHER

- Git
- GitLab
- Subversion
- Jira
- Confluence
- Kibana
- ANTLR
- JavaCC
- ZXing (QR)
- PayPal API
- RDF/XML/JSON
- SAP (BAPI)
- Phonegap
- JBoss
- Tomcat

DEVOPS / CLOUD

- Docker
- Kubernetes
- Rancher
- GitLab CI
- Jenkins
- Maven
- Grunt
- npm
- Artifactory
- AWS (EC2, S3, SageMaker, VPC)
- Microsoft Azure
- DigitalOcean

PERSISTENCE / DATABASES

- MySQL
- MongoDB
- PostgreSQL
- Oracle
- Neo4j
- JPA
- Hibernate

TESTING

- JUnit
- TDD
- Mockito
- Selenium
- DBUnit
- Playwright

DATA SCIENCE

- pandas
- numpy
- dplyr
- Jupyter
- RMarkdown
- ggplot
- seaborn
- Vega-Lite
- A/B Testing
- Webscraping

PROJECT HISTORY (SELECTION FROM 2014)

since 12/2025
ongoing

Autonomous AI agent systems — Harness & Agentic Engineering

Solytics GmbH · Remote — AI / SaaS

Harness engineering · Agent engineering · AI system architecture · Full-stack engineering · DevOps

Building and running autonomous LLM agent systems as a **harness engineer** — the engineering around the model, not the model itself. The harness is the product: the **agent loop** and its stop conditions, **tool use over MCP**, layered **memory** (durable facts, running context, session state), **approval gates** for anything irreversible, **quality gates** that must pass before an agent may ship, **observability** over every run, **budget and rate-limit control** with model routing and fallback, plus recovery when a run is killed mid-flight. A local multi-model proxy keeps the whole thing independent of any single provider. Two systems in production: a self-directing AI business partner that plans, writes code, tests and deploys around the clock, and **Fabrik** (fabrikhq.com), a multi-agent platform for autonomous software delivery with parallel worker agents. Event-driven orchestration over Telegram, GitHub and systemd; several AI-built SaaS products shipped end to end.

- Claude / Anthropic API
- LLM-Agenten
- Agent Harness
- Multi-Agent-Orchestration
- Tool Use / MCP
- RAG
- Prompt Engineering
- Python
- FastAPI
- Vue 3
- Go
- Docker
- systemd
- GitHub Actions
- DigitalOcean
- SQLite
- TDD
- TypeScript

07/2024–06/2025

12 months

Digital property profile for owners

verkaufe.immobilien · Remote — Internet

Full-stack engineering · Solution design · DevOps

Full-stack development of a digital property profile that owners create and maintain themselves: Vue.js front-end, Python back-end (Eve) with a REST API and MongoDB. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines for automated build and deployment. Worked to DevOps principles, owning build and operation of the services in Docker. Test-driven development, automated end-to-end tests with Playwright.

Python

Python Eve

MongoDB

TDD

Vue.js

Docker

GitLab

GitLab CI

REST

Playwright

07/2022–04/2024

22 months

Front- and back-end, heating cost billing

Brunata · Remote — Industry

Full-stack engineering · Solution design · DevOps

Production server-side services and APIs in Java and Spring Boot: a microservice that parses and generates heating-cost files in the Haweiko-Arge industry format. Designed and built in a microservices architecture with REST interfaces (OpenAPI, Swagger), secured with Spring Security and Keycloak, persisted in MongoDB. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines for automated build and deployment. Worked to DevOps principles, owning build and operation of the services in Docker. Test-driven development (JUnit, Mockito), automated end-to-end tests with Playwright. Front-end development.

Java

Spring Boot

MongoDB

JUnit

TDD

React

Docker

GitLab

GitLab CI

Spring Security

Keycloak

Mockito

OpenAPI

Swagger

REST

Microservices

Playwright

04/2021–11/2021

8 months

Software and data analysis in marketing

DigitalMarketer.com · Remote — Marketing

Full-stack engineering · Prototyping · Solution design · Data analysis

Server-side services in Java and Spring Boot: web-scraping microservices plus microsites (Vue.js, Quasar) with a Spring Boot back-end for calculators, surveys and questionnaire funnels; data analysis in R. Designed and built in a microservices architecture with REST interfaces (OpenAPI, Feign), persisted in PostgreSQL and MongoDB. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines. Worked to DevOps principles, owning build and operation of the services in Docker and Kubernetes. Test-driven development (JUnit, Mockito).

Scrum

Spring Boot

Docker

Kubernetes

Microservices

REST

R

Python

Java

OpenAPI

Feign

JUnit

TDD

Mockito

Vue.js

Quasar

MongoDB

dplyr

GitLab CI

GitLab

PostgreSQL

Python Eve

Python Flask

DigitalOcean

Webscraping

A/B Testing

Topic Modeling

01/2021–03/2021

3 months

Marketing campaign tooling

DigitalMarketer.com · Remote — Marketing

Full-stack engineering · Solution design

Tools that generate marketing campaigns and publish them to Google Ads and Twitter Ads: Java/Spring Boot microservices with REST interfaces (OpenAPI, Feign), Vue.js/Quasar front-end, storage in AWS S3. Designed and built in a microservices architecture. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines. Worked to DevOps principles, owning build and operation of the services in Docker and Kubernetes. Test-driven development (JUnit, Mockito).

Scrum

Spring Boot

Docker

Kubernetes

Microservices

REST

Java

OpenAPI

Feign

JUnit

TDD

Mockito

Vue.js

Quasar

AWS S3

GitLab CI

GitLab

12/2019–12/2020

13 months

Web-based configuration of telecommunications devices

Deutsche Telekom · Darmstadt/Remote — Telecommunications

DevOps · Build & configuration management · Full-stack engineering

Took over an existing prototype for configuring telecommunications devices over a web interface, developed it further and moved it into production. Production server-side services and APIs in Java and Spring Boot in a microservices architecture (REST, Feign, Spring Data/JPA), Vue.js front-end, secured with Keycloak, persisted in MySQL and MongoDB. Worked to DevOps principles, owning build, operation and monitoring of the services: Docker and Kubernetes (Rancher), logging and tracing via Kibana, Sleuth and Zipkin. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines for automated build and deployment. Test-driven development (JUnit, Mockito).



11/2018–03/2019

5 months

Image homogeneity of product niches from product photos

Sellics · Remote — E-commerce

Machine learning · Data analysis

A model that quantifies the image homogeneity of Amazon product niches to spot under-served niches early. Trained a ResNet50 CNN with transfer learning (Keras) and deployed it through a build pipeline as a Java microservice with DeepLearning4J into the existing landscape; prepared a Kafka data pipeline for continuous retraining.



06/2018–10/2018

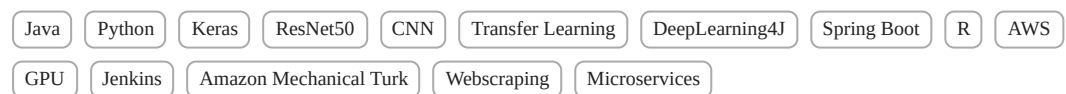
5 months

Aesthetic classification of Airbnb photos

AirDNA · Remote — Travel

Machine learning · Data analysis

A microservice that scores the quality of Airbnb photos: a Python scraper built an image database (~15k images), labelling via Amazon Mechanical Turk, exploratory data analysis in R; trained a CNN with transfer learning (Keras, ResNet50) on an AWS GPU instance and deployed it as a Spring Boot microservice with DeepLearning4J.



08/2017–03/2019

20 months

Connected Van — driver's logbook

Volkswagen · Remote — Automotive

Backend engineering

Microservices for vehicle geolocation and a fuel logbook.



07/2015–07/2017

25 months

Invoice management platform (SME)

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

Platform covering the full invoice life cycle (creation, payment, dunning, debt collection); a range of Spring Boot microservices.

Spring Boot

Docker

Kubernetes

Microservices

REST

Java

Maven

Spring Security

OAuth2

Spring Integration

MongoDB

Jira

Confluence

JUnit

TDD

Mockito

Selenium

AngularJS

Grunt

npm

Spring

Git

02/2015–06/2015

5 months

Invoice management platform (SME) — debt-collection integration

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

Microservice integrating a debt-collection provider

Spring Boot

Docker

Kubernetes

Microservices

REST

Java

Maven

Spring Security

OAuth2

Spring Integration

MongoDB

Jira

Confluence

JUnit

TDD

Mockito

Selenium

AngularJS

Grunt

npm

Spring

Git

09/2014–12/2014

4 months

Invoice management platform (SME) — prototype

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering · Solution design · Prototyping · Build & configuration management

Evaluation of microservices and SPA approaches, prototype for an invoice management platform.

Java

JavaScript

HTML

CSS

Spring

Bootstrap

AngularJS

Grunt

npm

Spring Boot

Docker

Kubernetes

Microservices

REST

Maven

Spring Security

MongoDB

Jira

Confluence

Git

CONTACT & CREDENTIALS

CONTACT & PROFILES

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PUBLICATION	Towards Data Science — author

EDUCATION

11/1997	Dipl.-Ing. (Univ.) Electrical Engineering (MSc equivalent) University of Erlangen-Nuremberg
05/1989	Abitur (university entrance qualification) Gymnasium am Romäusring, Villingen-Schwenningen

CERTIFICATES

12/2018	Machine Learning Engineer Nanodegree Udacity Verify →
08/2017	Data Analyst Nanodegree Udacity Verify →

